

Xuenan Li

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Research Interests

Calculus of Variations, Partial Differential Equations and Spectral Theory
with particular focus on problems from physics, mechanics, and materials science

New Methods for Designing Rigid Frameworks with higher-order rigidity
including an optimization-based approach to finding new tensegrity structures

Academic Position

2023- Chu Assistant Professor in Applied Mathematics
 COLUMBIA UNIVERSITY
 Supervisor: Michael I. Weinstein

Education

2018- 2023 Ph.D. in Mathematics
 COURANT INSTITUTE OF MATHEMATICAL SCIENCES, NEW YORK UNIVERSITY, NY, USA
 Advisor: Robert V. Kohn
 Thesis: Guest-Hutchinson modes and the macroscopic energy of the Kagome lattice metamaterial and related systems

2015-2018 B.S. in Mathematics and Data Science (double major)
 UNIVERSITY OF MICHIGAN, ANN ARBOR, MI, USA

2013-2015 B.S. in Mathematics
 SICHUAN UNIVERSITY, CHENGDU, CHINA

Publications & Preprints

Li, Xuenan and Kohn, Robert V. (2023) *Some results on the Guest-Hutchinson modes and periodic mechanisms of the Kagome lattice metamaterial*, J Mech Phys Solids 178 (Sept 2023) 105311, [published version](#), [arXiv:2210.00382](#) (55 pages).

Li, Xuenan, Mihnea Leonte, Santangelo, Christian D., and Holmes-Cerfon, Miranda (2025) *A Constrained Saddle Search Approach for Constructing Singular and Flexible Bar Frameworks*, accepted by 3rd IMA Conference on Mathematics of Robotics (peer reviewed), [arXiv:2503.14807](#) (9 pages).

Li, Xuenan and Kohn, Robert V. (2025) *The existence of effective energy of general lattice metamaterials*, in revision for *Journal of Elasticity*, [arXiv:2505.05436](#) (74 pages).

Li, Xuenan and Kohn, Robert V. (2025) *A nonlinear homogenization-based perspective on the soft modes and effective energies of some conformal metamaterials*, [arXiv:2509.16907](#) (54 pages).

Li, Xuenan, Santangelo, Christian D., and Holmes-Cerfon, Miranda. (2025) *A Constrained Optimization Approach for Constructing Rigid Bar Frameworks with Higher-order Rigidity*, [arXiv:2509.23072](#) (31 pages).

Li, Xuenan and Weinstein Michael I. (2025) *Pseudo-magnetism in the slowly-strained discrete honeycomb lattice*, in preparation.

Li, Xuenan, Liu, Chun, and Qi, Di. (2025) *An Asymptotic Approach for Modeling Multiscale Complex Fluids at the Fast Relaxation Limit*, in preparation.

Teaching Experience

Fall 2025	<i>Instructor</i> APAM E4204 Functions of a complex variable Columbia University
Spring 2025	<i>Instructor</i> APAM E4101 Dynamical systems Columbia University
Fall 2024	<i>Instructor</i> APAM E4204 Functions of a complex variable Columbia University
Spring 2024	<i>Instructor</i> APAM E4101 Dynamical systems Columbia University
Fall 2023	<i>Instructor</i> APAM E4204 Functions of a complex variable Columbia University
Fall 2022	<i>Teaching Assistant</i> MATH-UA.0262-007 Ordinary Differential Equations Courant Institute, New York University
Spring 2022	<i>Teaching Assistant</i> MATH-UA.0140-007: Linear Algebra Courant Institute, New York University
Fall 2021	<i>Teaching Assistant</i> MATH-UA.0263-002: Partial Differential Equations Courant Institute, New York University
Spring 2021	<i>Teaching Assistant</i> MATH-UA.0121-017: Calculus I Courant Institute, New York University

Conferences & Workshop Talks

- 09/2025 *A Constrained Saddle Search Approach for Constructing Singular and Flexible Frameworks*, 3rd IMA Conference on Mathematics of Robotics, Manchester, UK, Sep 25, 2025.
- 06/2025 *Edge states in the discrete slowly-varying Su–Schrieffer–Heeger (SSH) model*, SIAM Annual Meeting (AN25), Montreal, July 28, 2025.
- 06/2025 *Edge states in the discrete slowly-varying Su–Schrieffer–Heeger (SSH) model*, ETOPIM 13, New York, June 16, 2025.
- 05/2025 *A Constrained Optimization Approach for Constructing Rigid Bar Frameworks with Higher-Order Rigidity* (poster), NSF CompMath Meeting 2025, University of Utah, May 8-9, 2025.
- 05/2025 *A Constrained Optimization Approach for Constructing Bar Frameworks with Higher-order Rigidity*, Sayas Numerics Day 2025, University of Delaware, May 3, 2025.
- 04/2025 *A Constrained Optimization Approach for Constructing Tensegrity Frameworks*, East Coast Optimization Meeting 2025, April 18, 2025.
- 04/2025 *Deformations in soft mechanical metamaterials: modeling, analysis, and applications*, Workshop for Women in PDE and Applied Mathematics in Northeast Region, University of Connecticut, April 12, 2025.
- 04/2025 *A Constrained Optimization Approach for Constructing Singular Bar Frameworks*, ICERM Geometry of Materials Workshop, April 08, 2025.
- 04/2025 *Edge states in the discrete slowly-varying Su–Schrieffer–Heeger model*, AMS Spring Eastern Sectional Meeting, April 6, 2025.
- 06/2024 *The Kagome Lattice as a Mechanism-based Mechanical Metamaterial* (poster), Mathematics of Multiscale and Multiphysics Phenomena in Materials Science, Banff, Calgary, June 16 - 21, 2024.
- 05/2024 *The Kagome Lattice as a Mechanism-based Mechanical Metamaterial*, SIAM Conference on Mathematical Aspects of Materials Science (MS24), Pittsburgh, Pennsylvania, May 19–23, 2024.
- 10/2023 *The Kagome lattice as a mechanical metamaterial*, 2023 SES Annual Technical Meeting (SES 2023), Saint Paul, Minnesota, October 8-11, 2023.
- 8/2023 *The macroscopic behavior of the Kagome lattice metamaterial*, 10th International Congress on Industrial and Applied Mathematics (ICIAM 2023), Tokyo, August 20-25, 2023.
- 10/2022 *Some results on the Guest-Hutchinson modes and periodic mechanisms of the Kagome lattice metamaterial* (poster), Simons Collaboration on Extreme Wave Phenomena Based on Symmetries Annual Meeting, Flatiron Institute, New York, October 20–21, 2022.
- 10/2022 *The macroscopic energy of the Kagome lattice metamaterials*, UpState NY Soft Matter Symposium, Rochester Institute of Technology, October 11, 2022.
- 7/2022 *Kagome lattice as a mechanical metamaterial*, SIAM Annual Meeting (AN22), Pittsburgh, July 11-15, 2022.

- 6/2022 *The Kagome lattice as a mechanical metamaterial* (poster), UMass Summer School on Soft Solids and Complex Fluids, University of Massachusetts Amherst, June 5–9, 2022.
- 10/2021 *A mathematical perspective on Guest-Hutchinson modes* (poster), Simons Collaboration on Extreme Wave Phenomena Based on Symmetries Annual Meeting, Flatiron Institute, New York, October 21–22, 2021.

Invited Colloquium & Seminar Talks

- 03/2025 *Deformations in soft mechanical metamaterials: modeling, analysis, and applications*, Fluid Mechanics & Waves Seminar, NJIT, March 24, 2025.
- 11/2024 *Deformations in soft mechanical metamaterials: modeling, analysis, and applications*, Analysis & PDE Seminar, Worcester Polytechnic Institute, November 14, 2024
- 10/2024 *Deformations in soft mechanical metamaterials: modeling, analysis, and applications*, PDE & Differential Geometry Seminar, University of Connecticut, October 28, 2024
- 2/2024 *The Kagome lattice as a mechanical metamaterial*, PDE Seminar, Arizona State University, Phoenix, February 2, 2024.
- 11/2023 *The Kagome lattice as a mechanical metamaterial*, Data Science Seminar, University of Minnesota Twin Cities, Saint Paul, Minnesota, November 27, 2023.
- 11/2023 *The Kagome lattice as a mechanical metamaterial*, Applied Math Seminar, The University of British Columbia, Vancouver, November 7, 2023.
- 1/2023 *The Kagome Lattice as a Mechanism-based Metamaterial*, Applied Math Colloquium, Columbia University, New York, January 9, 2023.
- 3/2022 *Some results on Guest-Hutchinson modes and periodic mechanisms of the Kagome lattice as a metamaterial*, Modeling Simulation Group Seminar, Courant Institute, New York, March 3, 2022.
- 10/2021 *The Kagome lattice as a mechanical metamaterial*, Analysis and Applied Mathematics Seminar, University of Illinois Chicago, Chicago, October 11, 2021.

Academic Services

- Fall 2025 *Co-organizer*, Advances in the Spectral Analysis of Schrödinger Operators in Complex Materials SIAM Conference on Analysis of Partial Differential Equations (PD25) Pittsburgh, Pennsylvania, U.S., November 17–20, 2025
- Spring 2025 Thesis defense committee member:
PhD Candidate: *Edith Jin Zhang*, Department of Applied Physics and Applied Mathematics
Dissertation Title: *Ginzburg–Landau Functionals and Reaction–diffusion Equations in the Large-Graph Limit*.
- Fall 2024 – now Master Student Research Program at Columbia University with *Wendi Lu*.
- Fall 2024 Thesis defense committee member:
PhD Candidate: *Zirui Xu*, Department of Applied Physics and Applied Mathematics

Dissertation Title: *Ohta–Kawasaki Energy and Its Phase-Field Simulation.*

- Fall 2024 *Co-organizer*, APAM Research Conference, Columbia University, New York, NY
- Summer 2024 Undergraduate Student Research Program at Columbia University with two undergraduate students *Mihnea Leonte and Andrew Yang*;
Program title: *Designing singular structures in bar-joint networks*;
Funded by *Bonomi Scholars* at Columbia University.
- Fall 2025
update Mihnea Leonte received the Applied Mathematics Faculty Award of class 2025 and is a current Master student in Financial Engineering at Columbia University;
- Andrew Yang received the Applied Physics Faculty Award of class 2025 and is now a PhD student in Applied Physics at California Institute of Technology;
- Wendi Lu is now a current PhD student in Mathematics at University of Wisconsin-Madison.

Honors & Awards

- 2025 NSF-AWM travel grant
- 2023 SIAM Student Travel Award, 10th International Congress on Industrial and Applied Mathematics (ICIAM 2023)
- 2022 Sandra Bleistein Prize, New York University
- 2022 SIAM Student Travel Award, SIAM Annual Meeting (AN22)
- 2018–2023 New York University MacCracken Graduate Scholarship, New York University
- 2018 Wilfred Kaplan Award in Applied Mathematics, University of Michigan
- 2018 Outstanding Graduating Senior Award, University of Michigan
- 2017–2018 James B. Angell Scholar, University of Michigan
- 2017 Sumner B. Myers Award in Analysis, University of Michigan
This is awarded to the undergraduate student who is most excellent in the study of analysis.
- 2015 Chinese National Scholarship, Sichuan University
This scholarship is given to the top 3 students to honor their distinguished achievements.

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<https://xuenanli.github.io/>